Resume Seniority Classification Under Style and Demographic Bias (January 2026)

E. Fabrikant, R. Deeb.
Afeka Academic Collage of Engineering in Tel Aviv , Israel

Corresponding author: First A. Author (e-mail: fabrikant2002@gmail.com) , Second B. Author (e-mail: Ranedeeeb@gmail.com)

ABSTRACT Resume-based seniority inference is difficult due to heterogeneous structure and self-presentation that can inflate or downplay identical experience. We study three-class seniority classification (Junior/Mid/Senior) under accuracy, style robustness, and demographic sensitivity using controlled counterfactual edits. From 4,817 resumes, we build a leakage-resistant pipeline and a class-balanced subset of 2,100 profiles. We benchmark TF–IDF + Random Forest, fine-tuned transformer encoders (DistilBERT, RoBERTa), and zero-shot foundation models (GPT-5, Gemini 3 Pro, Claude Sonnet 4.5). In the standard setting (Test 1), supervised models achieve the highest reliability (≈0.70–0.73 accuracy/macro-F1), while the lexical baseline is competitive but relies strongly on numeric shortcuts (e.g., total years of experience); zero-shot LLMs yield substantially lower macro-F1 and a systematic upward shift (Junior to Mid, Mid to Senior). In controlled style triplets (Test 2), fine-tuned encoders exhibit high instability across neutral/ understated / overstated variants (up to ~93% label changes), whereas foundation models are markedly more tone-stable. Under demographic name swaps (Test 3), group-level differences are generally small relative to style effects, with limited evidence of race-coded shifts and modest gender-coded variation in some models. Overall, accuracy alone is insufficient: deployment-relevant evaluation requires leakage-aware benchmarks plus controlled robustness and fairness tests.

INDEX TERMS bias analysis, data leakage prevention, fine-tuning, foundation models, fairness, natural language processing (NLP), resume parsing, seniority classification, text classification, transformers, zero-shot learning, GenerativeAI

I. INTRODUCTION

Automated resume screening is increasingly used in hiring workflows, making reliability, robustness, and fairness critical requirements. Inferring a candidate's seniority level (Junior/Mid/Senior) from resume text is challenging due to heterogeneous formatting, noisy and lengthy content, and self-presentation strategies that may inflate or downplay identical factual experience. Misclassification can lead to systematic under- or over-evaluation of candidates and, if model behavior is sensitive to demographic cues, may amplify existing disparities. Prior work has shown that seniority classification is feasible and that fine-tuned transformer encoders often outperform classical baselines and zero-shot prompting, but important questions remain regarding stability under style manipulations and sensitivity to demographic proxies in modern model generations. In this study, we evaluate seniority inference using three complementary evaluations: (1) standard held-out classification performance, (2) robustness under controlled style triplets (neutral/overstated/understated) that preserve factual content, and (3) demographic sensitivity under controlled name swaps that alter gender- and race-coded signals while keeping all resume content constant.

We begin with 4,817 resumes and construct a leakage-resistant preprocessing and labeling pipeline, including normalization of key sections and removal of fields that could trivially reveal the target label. To enable fair comparison across classes, we create a class-balanced subset of 2,100 resumes (700 per class). We then benchmark three modeling paradigms: a TF–IDF + Random Forest baseline, fine-tuned transformer classifiers (DistilBERT and RoBERTa), and zero-shot foundation models (GPT-5, Gemini 3 Pro, and Claude Sonnet 4.5). Beyond Test 1, we measure deployment-relevant failure modes using explicit stability metrics (inconsistency and rank shift) on style triplets (Test 2) and group-difference indicators under demographic name swaps, including intersectional comparisons (Test 3).

Contributions:

- Leakage-aware benchmark for seniority classification: We construct a reproducible labeling and preprocessing pipeline and evaluate on a class-balanced subset of 2,100 resumes.

- Comparison across classical, fine-tuned, and zero-shot paradigms: We benchmark TF–IDF + Random Forest, fine-tuned transformer encoders (DistilBERT, RoBERTa), and recent foundation models under a unified evaluation protocol.
- Controlled robustness and fairness evaluation: We quantify prediction stability under style rewrites (neutral/overstated/understated) and sensitivity to demographic name cues via counterfactual swaps, including intersectional effects.

Research Questions:

1. Performance: Which modeling paradigm is most reliable on standard held-out seniority classification?
2. Style robustness: To what extent do predictions change when factual content is fixed but tone varies (neutral vs. overstated vs. understated)?
3. Demographic sensitivity: Do seniority predictions shift under controlled demographic name swaps, and are any effects amplified under intersectional conditions?

Paper Organization

The remainder of this paper is organized as follows: Section II reviews related work, Section III describes the dataset and methodology, Section IV presents results, and Section V discusses implications and limitations. Code and materials are available at the project repository on GitHub. https://github.com/AFKRainn/SeniortyPrediction

II. RELATED WORK

Prior work has shown that seniority classification from resume text is feasible but sensitive to self-presentation. Cohen et al. [1] introduced a triplet-based evaluation in which each profile appears in Normal / Overstated / Understated versions that preserve factual content while varying rhetorical style. Their results indicate that fine-tuned transformer encoders outperform sparse baselines and zero-shot prompting, suggesting that supervised adaptation helps capture seniority-relevant cues beyond bag-of-words features. However, their study leaves open questions about formal stability metrics across triplets, demographic fairness beyond aggregate performance, and which linguistic signals systematically drive level decisions. These gaps motivate our use of explicit stability measurements and controlled counterfactual evaluations. LLMs are increasingly used in HR pipelines beyond classification. Cohen and Levi [2] describe an applied resume job matching system that extracts skills and generates fit analyses, illustrating practical deployment of LLM-based resume understanding. Complementarily, the NAACL Industry study [3] evaluates resume job matching under controlled counterfactual changes (e.g., names signaling gender/race and educational institutions), showing that performance improvements do not necessarily imply fairness improvements; notably, institutional prestige effects can persist even when other biases decrease. While these works focus on matching rather than seniority classification and do not study style manipulations, their methodology of changing one factor at a time directly informs our controlled design.

Bias in hiring has been extensively documented in both human and automated screening. Derous and Ryan [4] propose a framework explaining how identity cues and ambiguity shape impressions and downstream decisions, suggesting that judgments may be influenced by signals beyond job-relevant facts. Recent work further emphasizes that biases can be intersectional and unevenly distributed across groups [5], and that models may implicitly weight proxy signals such as reputation or company prestige even when overall accuracy is high [6]. These findings support evaluating not only average performance but also group-level shifts under controlled demographic cues. Finally, research on manipulation and self-presentation provides a behavioral and mechanistic basis for style-based evaluation. Studies on deception detection indicate that dedicated fine-tuning can outperform general prompting when the goal is robustness to subtle linguistic manipulation [7]. Separately, evidence from hiring contexts shows that self-glorification can increase acceptance rates while modest language may be interpreted as lower competence, with potential gender-correlated differences in style [8]. Together, this literature supports the need for seniority classifiers that are robust to rhetorical inflation/understatement and do not systematically advantage more assertive presentation.

In summary, existing work motivates three requirements for modern seniority classification: (1) competitive accuracy on real resume distributions, (2) robustness to controlled style variation, and (3) fairness evaluation under controlled demographic proxies, including intersectional effects. Our study addresses these requirements by benchmarking classical and transformer-based models alongside modern foundation models, and by evaluating stability and group-level shifts under controlled counterfactual edits.

III. METHODOLOGY

This section describes the methodology used to build and evaluate a three-class resume seniority classifier (Junior/Mid/Senior). We outline data cleaning and labeling, the rule-based construction of a consistent "last relevant role" target with leakage controls, and enriched summary generation to improve input quality. We then define the resume-to-text representation and benchmark three modeling paradigms: a TF–IDF + Random Forest baseline, fine-tuned transformer encoders, and zero-shot foundation models. Evaluation is conducted under three complementary settings: Test 1 (standard held-out performance), Test 2 (controlled style triplets: neutral/overstated/understated with identical facts) to quantify stability under self-presentation changes, and Test 3 (controlled demographic name swaps) to measure group-level and intersectional prediction shifts under fixed

content. An overview of the end-to-end pipeline is shown in Fig. 1.

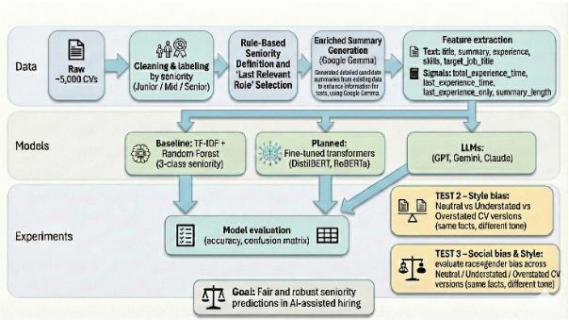

FIGURE 1. Summarizes our end-to-end pipeline: we start from raw resumes, apply cleaning and rule-based labeling to derive seniority ground truth, generate enriched summaries for specific experiments, extract text features and auxiliary signals, and evaluate three modeling paradigms (baseline ML, fine-tuned transformers, and zero-shot LLMs) under standard performance metrics and controlled robustness/fairness tests.

A. DATASET OVERVIEW

We start from master_resumes.jsonl [9] which contains 4,817 resume profiles with structured fields (e.g., experience entries, education, skills) and free-text components (e.g., summaries and role descriptions). We evaluate models using three datasets aligned with Tests 1-3. Test 1 (standard performance): after cleaning, labeling, and leakage control (Sections B-C), we construct a class-balanced subset of 2,100 resumes (700 per class) used for supervised training and held-out evaluation. Test 2 (style robustness): to isolate sensitivity to self-presentation, we sample 120 base resumes evenly distributed across classes (40/40/40) and generate three stylistic variants per resume (neutral/ overstated/ understated), yielding 360 resumes used only for controlled robustness evaluation. Test 3 (demographic sensitivity): To measure demographic cue effects under fixed content, we take the 120 neutral resumes from Test 2 and create four name-coded demographic variants per resume (race × gender), yielding 480 resumes used only for controlled fairness evaluation.

B. CLEANING & LABELING

Seniority labels are derived from the dataset's experience-level annotations (the level field within experience entries). We map the original taxonomy into a unified three-class target (Junior/Mid/Senior) to enable consistent comparison across models. The raw dataset is class-imbalanced; therefore, for Test 1 we construct a class-balanced subset (Section A) to avoid performance inflation toward majority classes. Resumes are additionally normalized to reduce formatting noise prior to modeling. We apply standard text cleaning (e.g., whitespace and punctuation normalization and consistent casing) and normalize structured fields into a consistent representation (e.g., flattening semi-structured lists into comparable text segments). This preprocessing reduces superficial variation so that downstream evaluation emphasizes content rather than formatting artifacts.

C. RULE-BASED SENIORITY DEFINITION AND 'LAST RELEVANT ROLE' SELECTION

To assign a single, consistent target label per candidate, we apply a rule-based "last relevant role" selection. If a profile contains a single experience entry, we use its associated level annotation. If multiple roles are listed, we select the most recent role (or the current role when an end date is missing) and map its level into the unified three-class scheme (Junior/Mid/Senior). This produces one seniority target per resume for downstream training and evaluation.To verify that the resulting labels are not reducible to a simple numeric shortcut, we compute total years of experience from employment timelines and examine its distribution across classes (Fig. 2). While junior profiles concentrate at lower ranges and senior profiles at higher ranges, substantial overlap remains, indicating that experience duration alone does not cleanly separate the labels. This motivates relying on richer resume signals (role descriptions, responsibilities, skills, and projects) rather than tenure alone. Leakage prevention: To prevent trivial label prediction, we remove or mask fields that directly encode seniority before model training and evaluation, including the per-position level attribute used for labeling and any derived fields that mirror the target label. Model inputs are constructed only from resume content fields described in Section E.

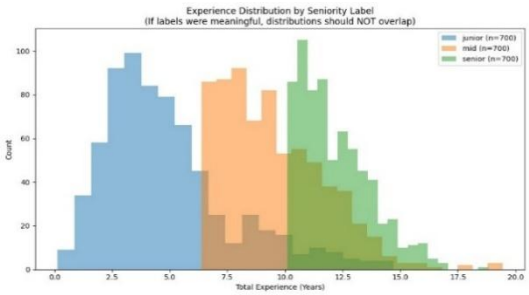

FIGURE 2. Total years of experience distribution by seniority label (Junior/Mid/Senior). The three distributions overlap, indicating that experience alone does not perfectly separate the classes.

D. ENRICHED SUMMARY GENERATION

Manual inspection indicated that the dataset's original summary field was frequently repetitive and weakly aligned with the underlying resume content. To improve input quality and consistency across models, we generate a new summary for each profile using only structured resume fields (e.g., experience, skills, projects). We compare two LLMs Google's Gemma and Mistral 3B against the original summaries using two criteria. Length control: we target a concise profile summary; an industry resume-trends report estimates an average profile length of ~55 words [10]. Gemma produces summaries of ~60 words on average, while Mistral 3B

averages ~170 words; therefore, we select Gemma. Content grounding (semantic alignment): to verify that summaries reflect the resume rather than generic boilerplate, we measure semantic similarity between each summary S_i and a resume reference text R_i formed by concatenating key sections (experience, skills, projects). We embed both texts using SentenceTransformer (all-MiniLM-L6-v2) and compute cosine similarity. Generated summaries achieve substantially higher alignment than the original summaries (mean similarity 0.660 vs. 0.350), indicating stronger grounding in candidate-specific content. We use the selected enriched summaries as part of the model input across Tests 1-3.

E. FEATURE EXTRACTION

All model families operate on a unified resume-to-text representation constructed from the same content fields to ensure comparability. For each candidate, we form a single input document by concatenating: the enriched summary (Section D), experience (job titles and role descriptions ordered chronologically with the oldest role first), skills (flattened into a delimiter-separated list), education (degree and institution text as provided), and projects/achievements when available. Fields that directly encode the target label (e.g., the position-level level) are excluded (Section C). To preserve structure, each component is preceded by a section tag (e.g., [SUMMARY], [EXPERIENCE], [SKILLS], [EDUCATION], [PROJECTS]), and list-valued fields are converted into a consistent textual format. For transformer encoders, the resulting text is tokenized and truncated to the model's maximum sequence length; for the TF–IDF baseline, the full concatenated text is used. The same input construction is applied in Test 1 and in the controlled evaluations: in Test 2, only the phrasing is rewritten while factual content is held constant. In Test 3, only the candidate name string is swapped while the remainder of the constructed input is unchanged.

F. MODEL FAMILIES

We evaluate three modeling paradigms for three-class resume seniority prediction (Junior/Mid/Senior): a sparse lexical baseline (TF–IDF + Random Forest) that relies on surface term statistics, fine-tuned Transformer encoders (DistilBERT, RoBERTa) that adapt contextual representations to our labeled dataset, and modern foundation LLMs (GPT-5, Gemini 3 Pro, Claude Sonnet 4.5) evaluated as prompt-only (zero-shot) classifiers. All models operate over the same label space and are evaluated under the same datasets and controlled protocols (Tests 1-3), enabling direct comparisons of accuracy, style robustness, and demographic sensitivity.

F.1 Baseline: TF-IDF + Random Forest

As a fast and interpretable baseline, each resume is converted into a sparse vector using TF-IDF weighting (bag-of-words with inverse-document down-weighting).View scikit-learn's TfidfVectorizer [11]. A Random Forest classifier is trained on these vectors to predict the seniority label [12]. This baseline is computationally efficient and provides a reference for how far keyword-driven methods can go without contextual understanding. However, because TF–IDF discards word order and long-range dependencies, it is naturally sensitive to synonyms, embellishment, and other stylistic variations motivating the controlled style tests in Test 2.

F.2 Fine-Tuned Transformers: DistilBERT and RoBERTa

We fine-tune two encoder-only Transformer backbones-DistilBERT [13] and RoBERTa [14] for supervised three-class classification. DistilBERT provides an efficiency-oriented model distilled from BERT, while RoBERTa is a higher-capacity encoder trained with an optimized pretraining recipe. Both models encode each resume into contextual token representations using self-attention, enabling integration of signals distributed across the document (e.g., scope, ownership, impact, and role responsibilities), rather than relying on isolated keywords. For encoder-style classification, the final representation of the special [CLS] token (denoted h_{CLS}) is mapped to logits:

$$z = Wh_{CLS} + b \mid \quad z \in R^3 \quad (1)$$

Class probabilities are obtained using SoftMax:

$$p(y = k|x) = \frac{e^{z_k}}{\sum_{j=1}^{3} e^{z_j}} \quad (2)$$

And training minimizes standard multiclass cross-entropy. Model selection is performed using validation performance (macro-F1) to favor balanced performance across classes. Importantly, the fine-tuned models are trained only on Test 1; in Tests 2-3, they are evaluated in inference mode only on controlled variants (style and demographic), ensuring robustness and fairness measurements reflect model behavior rather than additional adaptation. Fine-tuning updates both encoder and head parameters via backpropagation to minimize cross-entropy. Training hyperparameters and selection criteria are reported in Section G.

F.3 Foundation LLMs as Classifiers: Zero-Shot Protocol

Evaluates GPT-5 [15], Gemini 3 Pro [16], and Claude Sonnet 4.5 [17] as prompt-only classifiers in a strict zero-shot setting. Each model receives the same resume input format and a single instruction to output exactly one label from {Junior, Mid, Senior}. We use zero-shot (rather than few-shot) to avoid demonstration-selection effects and to preserve comparability across providers and across controlled evaluations. Since foundation LLMs may overweight persuasive phrasing or prestige cues, we evaluate them not only on Test 1 accuracy but also on Test 2 (neutral / overstated/ understated triplets

with identical facts) and Test 3 (demographic cue variants under fixed content) to quantify style sensitivity and demographic-related shifts under controlled conditions.

G. MODEL EVALUATION

We evaluate all approaches on three settings aligned with Tests 1-3. Test 1 measures standard predictive performance on held-out resumes. Tests 2-3 measure sensitivity under controlled perturbations where resume facts are held constant and only style (Test 2) or demographic cues (Test 3) are varied.

G.1 Test 1: Standard classification performance

We report, predict, and make confusion matrices to characterize overall performance and per-class error patterns. Let $C \in \mathbb{R}^{K\times K}$denote the confusion matrix where C_{ij}counts examples with true class $i$predicted as class $j(K = 3)$. Accuracy is:

$$Accuracy = \frac{\sum_{k=1}^{K} M_{kk}}{\sum_{i=1}^{K}\sum_{j=1}^{K} M_{ij}} \quad (3)$$

To equally weight all classes, we emphasize macro-F1, computed as the unweighted mean of per-class F1 scores:

$$F1_k = 2 \times \frac{\mathrm{P}_k \times \mathrm{R}_k}{\mathrm{P}_k + \mathrm{R}_k} \quad (4)$$

Finally, the Macro F1-Score is the average over all K classes:

$$\text{Macro F1} = \frac{1}{K}\sum_{k=1}^{K} F1_k \quad (5)$$

where $\boldsymbol{P_k}$and $\boldsymbol{R_k}$are per-class precision and recall (computed from the confusion matrix). Macro-F1 is appropriate for our balanced three-class setting and discourages majority-class bias.

G.2 Zero-shot LLM evaluation protocol (reproducibility)

For foundation LLMs, each resume is provided in the same structured text format, and the model is instructed to output exactly one label from {Junior, Mid, Senior}. We use deterministic decoding (fixed settings) and normalize outputs to the label set; responses outside the label set are treated as invalid and handled consistently across models (e.g., mapped via strict rules or counted as errors).

G.3 Test 2: Style robustness (controlled triplets)

To quantify sensitivity to self-presentation style while holding factual content constant, each base resume $i \in \{1, \dots, N\}$is evaluated in three stylistic variants $v \in$ {neutral, overstated, understated}, producing predictions $\hat{y}_{i,v}$. We report a Triplet Consistency Rate (TCR), defined as the fraction of resumes whose predicted label remains identical across all three variants:

$$TCR = \frac{1}{N}\sum_{I=1}^{N} 1[\hat{y}_i, netural = \hat{y}_i, overstated = \hat{y}_i, understated] \quad (6)$$

and the corresponding Inconsistency Rate IR $= 1 -$ TCR. To capture *directional* sensitivity (upgrading vs. downgrading), we use an ordinal mapping $r(\text{Junior}) = 1$, $r(\text{Mid}) = 2$, $r(\text{Senior}) = 3$and define the style-induced rank shift relative to neutral as:

$$\Delta_{i,v}^{style} = r(\hat{y}_{i,v}) - r(\hat{y}_{i,neutral}) \quad (7)$$

We summarize the rate of positive shifts under the overstated variant, negative shifts under the understated variant, and the mean shift magnitude, optionally stratified by the true class.

G.4 Test 3: Demographic sensitivity (controlled demographic variants)

To measure demographic cue effects under fixed resume content, we generate demographic name-coded variants $g \in G$ (race × gender cues) for each base resume and obtain predictions $\hat{y}_{i,g}$. Using the same ordinal mapping $r(\cdot)$, we define the demographic-induced shift relative to the neutral/name-default version as:

$$\Delta_{i,g}^{demo} = r(\hat{y}_{i,g}) - r(\hat{y}_{i,neutral}) \quad (8)$$

We report the overall flip rate (fraction with $\Delta_{i,g}^{demo} \neq 0$), the mean shift per demographic group, and a worst-group statistic (e.g., maximum absolute mean shift across g) to capture intersectional cases where specific groups experience larger systematic upgrading or downgrading despite identical qualifications.

H. ROBUSTNESS AND FAIRNESS EVALUATION (CONTROLLED TESTS)

Beyond standard held-out evaluation (Test 1), we conduct two controlled experiments designed to isolate model sensitivity to factors that should not affect seniority inference when candidate qualifications are unchanged. In both tests, each base resume serves as the content anchor, and only a single controlled variable is modified. Evaluation is reported using the robustness and shift metrics defined in Section G (Accuracy, Rank Difference, and consistency/inconsistency measures), with additional bias indicators for Test 3.

H.1 Test 2: Style Robustness (Controlled Triplets)

To measure sensitivity to self-presentation, we sample 120 base resumes evenly across seniority classes (40 Junior, 40 Mid, 40 Senior) and generate three stylistic versions per resume: neutral, overstated, and understated, while preserving factual content. This yields 360 resumes used only for robustness analysis. We quantify whether predictions remain stable across the three variants (consistency/ inconsistency) and whether models exhibit systematic upward shifts under overstated phrasing or downward shifts under understated phrasing, as reflected by rank-based shift statistics.

H.2 Test 3: Demographic Sensitivity (Controlled Name Variants)

To test for demographic cue effects under fixed qualifications, we take the 120 neutral resumes from Test 2 and create four name-coded demographic variants per resume (race × gender): Caucasian Male, Caucasian Female, African American Male, and African American Female, producing 480 resumes used only for fairness analysis. We measure prediction stability across demographic variants (inconsistency), directional rank shifts relative to the neutral baseline, and report three bias indicators: Racial Bias (Caucasian-coded vs. African American-coded), Gender Bias (male-coded vs. female-coded), and an Extreme Comparison (Caucasian Male vs. African American Female) to capture potential intersectional effects.

IV. RESULTS AND EXPERIENCES

A. OVERALL MODEL COMPARISON TEST 1

Table I summarizes Test 1 performance across all evaluated models using Accuracy, Macro-F1, and per-class F1 (Junior/Mid/Senior). Overall, the supervised models outperform the zero-shot LLMs in both accuracy and class balance. The TF–IDF + Random Forest baseline achieves the strongest overall performance (Accuracy = 0.726, Macro-F1 = 0.730), while the fine-tuned Transformer encoders perform similarly in Accuracy but slightly lower in Macro-F1 (DistilBERT: 0.726 / 0.720, RoBERTa: 0.707 / 0.700). A consistent pattern across supervised approaches is that Mid is the hardest class: Mid-F1 is substantially lower than Junior-F1 and Senior-F1 (e.g., 0.59 for the baseline, 0.55 for DistilBERT, 0.52 for RoBERTa), indicating overlap in the Mid/Senior boundary and suggesting that many profiles contain mixed signals (e.g., partial leadership/ownership cues but not consistently senior scope). This is also visible in the confusion matrices (Appendix B), where the dominant error mode is Mid → Senior. In contrast, the zero-shot LLMs underperform the supervised models and show weaker class balance, with notably lower Macro-F1 (GPT-5-mini: 0.402, Claude Haiku 4.5: 0.385, Gemini-3-flash (preview): 0.435). Within the LLM group, Gemini-3-flash yields the best overall balance (Accuracy = 0.500, Macro-F1 = 0.435), driven by stronger Mid/Senior separation (Mid-F1 = 0.49, Senior-F1 = 0.69). GPT-5-mini and Claude Haiku 4.5 exhibit a clearer upward shift tendency toward higher predicted seniority, reflected in weak Mid-F1 (0.24 and 0.11) alongside relatively higher Senior-F1 (0.67 and 0.63).

Model	Accuracy	Macro F1	Junior F1	Mid F1	Senior F1
TF-IDF Baseline	**0.726**	**0.730**	0.840	**0.590**	**0.740**
DistilBERT	**0.726**	0.720	**0.90**	0.550	0.720
RoBERTa	0.707	0.700	0.870	0.520	0.710
GPT-5	0.459	0.402	0.230	0.240	0.670
Claude Sonnet 4.5	0.450	0.385	0.420	0.110	0.630
Gemini 3 pro	0.500	0.435	0.130	0.490	0.690

TABLE 1. Overall performance comparison across evaluated models (Accuracy, Macro-F1, and per-class F1)

Supervised models better capture dataset-specific decision boundaries. Zero-shot LLMs show weaker calibration for early-career classes and tend to compress Junior/Mid upward.

B. *STYLE ROBUSTNESS UNDER CONTROLLED TRIPLES (TEST 2)*

Test 2 evaluates robustness to self-presentation style while holding qualifications constant. For each candidate, we generate three stylistic variants (Neutral/ Overstated/ Understated) and measure accuracy by style and prediction stability across the triplet using the Triplet Consistency Rate (TCR) (equivalently, inconsistency rate = 1 − TCR), producing 3N inputs from N base candidates (N = 120). Accuracy by style (Fig. 3) reports accuracy across the three styles. DistilBERT achieves 0.44 (Neutral), 0.33 (Overstated), and 0.45 (Understated), while RoBERTa achieves 0.36, 0.33, and 0.49, respectively. Both models degrade under the Overstated variant, indicating that confident/embellished phrasing can shift the decision boundary even when factual content is unchanged.

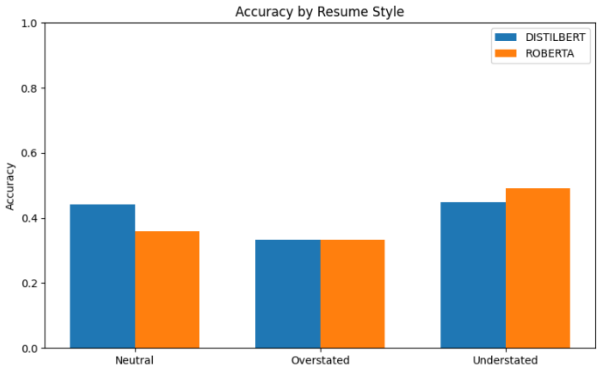

FIGURE 3. Accuracy by resume style (Neutral/ Overstated/ Understated) for fine-tuned models on Test 2.

Stability across triplets (Fig. 4) summarizes prediction inconsistency across the three variants (lower is better). DistilBERT changes its predicted label for 40.8% of candidates across the triplet, whereas RoBERTa changes its predicted label for 93.3% of candidates. This indicates that RoBERTa's inferred seniority is highly tone-dependent, and that rewriting style alone can cause large-scale label flips even when candidate facts are fixed.

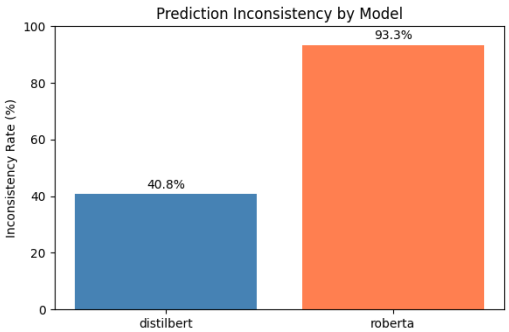

FIGURE 4. Prediction inconsistency rate across style triplets (Test 2); lower indicates higher triplet consistency.

In addition to overall instability, we examine whether style edits induce systematic directional shifts in predicted seniority. (Fig. 5) reports prediction inconsistency rates for the evaluated LLMs. GPT-5 exhibits the highest instability (32.5%), followed by Gemini 3 (16.7%), while Sonnet 4.5 remains highly stable (4.2%). These results indicate substantial differences in robustness to stylistic perturbations across models, with GPT-5 being notably more sensitive to rewriting than the other LLMs. Beyond flip frequency, we analyze directional bias using the average ordinal rank difference relative to ground truth, mapping {Junior=0, Mid=1, Senior=2}. As shown in Fig. Y, GPT-5 systematically overestimates seniority across all styles, with the strongest inflation under the Overstated variant (+0.408), compared to Neutral (+0.283) and Understated (+0.108). Gemini 3 shows moderate upward bias (Neutral +0.210, Overstated +0.218, Understated +0.075), while Sonnet 4.5 remains near zero across styles (Neutral +0.008, Overstated +0.000, Understated -0.017). These findings suggest that Overstated phrasing tends to inflate predicted seniority, particularly for GPT-5, whereas Understated phrasing reduces inflation magnitude. Most shifts occur between adjacent classes, especially around the Mid/Senior boundary, indicating that stylistic emphasis can tip the model across ordinal decision thresholds without changes in qualifications.

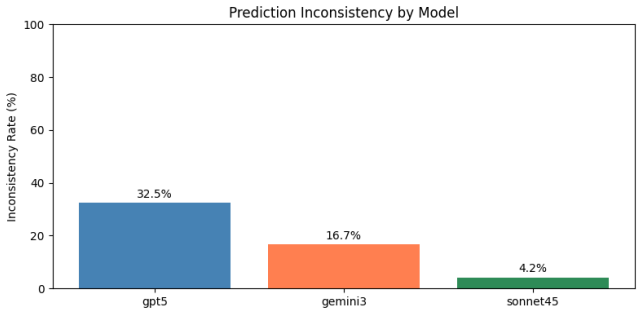

FIGURE 5. Prediction inconsistency rate across style-controlled triplets (Test 2) for zero-shot LLMs. Values represent the percentage of candidates whose predicted seniority changes across Neutral, Overstated, and Understated variants. Lower values indicate stronger robustness to stylistic perturbations.

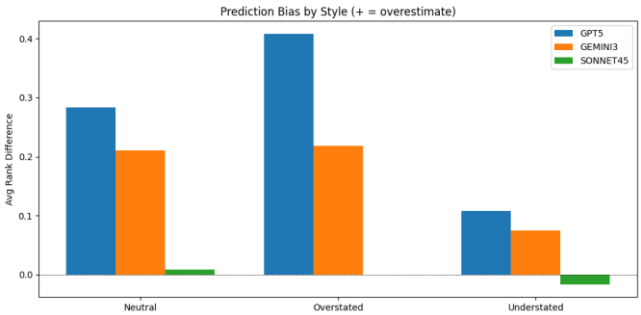

FIGURE 6. Average ordinal rank difference by resume style (Test 2) for zero-shot LLMs, where positive values indicate seniority overestimation. Ordinal mapping: Junior=0, Mid=1, Senior=2. Overstated phrasing induces stronger upward bias for GPT-5 and Gemini 3, while Sonnet 4.5 remains near zero across styles.

Overall, Test 2 demonstrates substantial robustness differences across LLMs, with Sonnet 4.5 exhibiting the strongest calibration and stability under controlled style edits.

C. *DEMOGRAPHIC SENSITIVITY UNDER NAME-ONLY VARIANTS (TEST 3)*

Test 3 isolates demographic cue effects by swapping name-coded race × gender cues while holding resume content fixed. For each resume $\boldsymbol{i}$and demographic variant $\boldsymbol{g}$, we compute the ordinal shift relative to the neutral prediction:

$$\Delta_{i,g} = ord(\hat{y}_{i,g}) - ord(\hat{y}_{i,neutral}) \quad (9)$$

where $\boldsymbol{ord}$ (Junior=0,Mid=1,Senior=2). We report group-level mean shifts and evaluate an extreme intersectional contrast (Caucasian Male vs African American Female). Additional subgroup breakdowns (race-only and gender-only, for all models) are provided in Appendix C. LLMs (race-coded variants). As shown in (Fig. 7), GPT-5 and Gemini 3 exhibit modest positive mean shifts across race-coded variants (upward seniority bias), while Sonnet 4.5 remains close to zero shift.

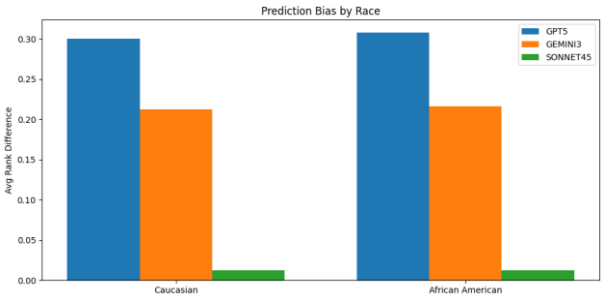

FIGURE 7. Mean ordinal rank shift Δunder race-coded name variants for zero-shot LLMs (positive values indicate upward seniority shifts).

Importantly, race-conditioned differences are small relative to each model's overall calibration tendency, indicating limited sensitivity to name-only race cues in this setup. We further stress-test demographic effects by comparing Caucasian Male vs African American Female name variants. For the fine-tuned transformers (Fig. 8), the two groups are nearly indistinguishable, suggesting that the dominant behavior is global overestimation rather than demographic-specific directional bias.

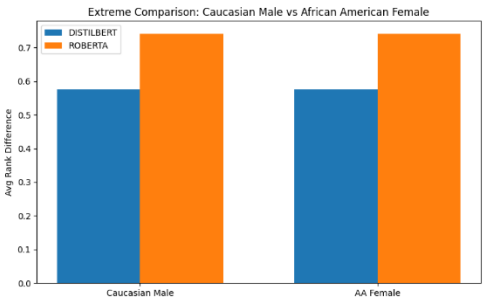

FIGURE. 8. Extreme intersectional comparison for fine-tuned transformers: mean ordinal rank shift Δfor Caucasian Male vs African-American Female name variants.

For the zero-shot LLMs (Fig. 9), GPT-5 shows the largest gap between the two groups, but the difference remains small compared to GPT-5's overall upward shift. Gemini 3 shows a smaller and more uniform shift, and Sonnet 4.5 remains essentially invariant.

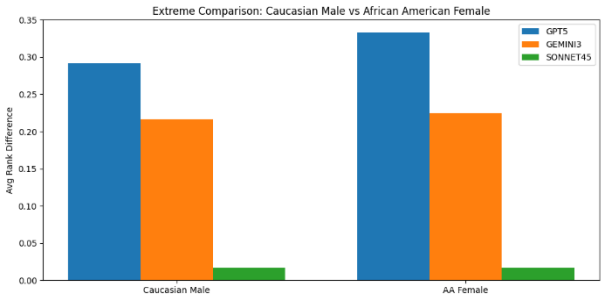

FIGURE. 9. Extreme intersectional comparison for zero-shot LLMs: mean ordinal rank shift Δ for Caucasian Male vs African-American Female name variants.

Overall, Test 3 suggests that name-coded demographic cues have a substantially smaller impact than controlled style edits (Test 2). Observed shifts are dominated by model-family calibration tendencies and provider differences rather than large systematic demographic-driven reclassification, see Appendix B (mean-shift subgroup plots) and Appendix C (subgroup accuracy plots).

V. DISCUSSION

A. *SUPERVISED vs. ZERO-SHOT PERFORMANCE AND ERROR MODES (TEST 1)*

Across Test 1, supervised models consistently outperform zero-shot foundation LLMs on the three-class seniority task. The TF–IDF + Random Forest baseline achieves strong performance (Accuracy = 0.726, Macro-F1 = 0.730), but removing the explicit "total years of experience" feature causes a sharp accuracy drop (to approximately 0.55), indicating strong reliance on an easily exploitable duration shortcut rather than deeper semantic cues. In contrast, the fine-tuned transformer encoders maintain competitive performance (DistilBERT: Accuracy = 0.726, Macro-F1 = 0.720; RoBERTa: Accuracy = 0.707, Macro-F1 = 0.700) and are less dependent on explicit experience-duration signals, suggesting they learn broader textual patterns (titles, responsibilities, scope, skills, leadership phrasing) that correlate with seniority. Zero-shot LLMs underperform relative to supervised approaches and are less aligned with the dataset's labeling scheme under prompt-only evaluation (e.g., lower Macro-F1 and weaker class balance), indicating that task-specific supervision remains important for reliable performance in this setup. A consistent error mode across model families is the Mid-level boundary, which is the hardest class to learn. Mid-F1 is systematically lower than Junior-F1 and Senior-F1, and errors are dominated by Mid → Senior confusion (as also reflected in the confusion matrices in Appendix B). This pattern suggests that "Mid" is inherently ambiguous and often contains mixed signals—some resumes express senior-like ownership or impact language without consistent scope, while others describe strong technical contributions without leadership or decision-making context- creating a blurred decision threshold between adjacent levels.

B. STYLE ROBUSTNESS UNDER CONTROLLED REWRITING (TEST2)

Test 2 shows that self-presentation style is a major confound for seniority prediction, even when qualifications are held constant. Accuracy degrades under stylistic inflation: both fine-tuned models perform worst on the Overstated variant (DistilBERT: 0.33; RoBERTa: 0.33), compared to higher accuracy on Neutral and Understated variants (Fig. 3). More importantly, controlled rewriting produces substantial prediction instability across style triplets. DistilBERT changes its predicted label for 40.8% of candidates across the three variants, while RoBERTa changes its label for 93.3% (Fig. 4), indicating that RoBERTa's inferred seniority is highly dependent on tone rather than stable evidence of scope/impact.For zero-shot LLMs, robustness is highly provider-dependent. Style triplets produce an inconsistency rate of 32.5% for GPT-5, 16.7% for Gemini 3, and only 4.2% for Sonnet 4.5 (Fig. 5), showing that some LLMs are far more stable under stylistic perturbations than others. Beyond flip frequency, directional shift analysis shows that Overstated phrasing systematically inflates predicted seniority—especially for GPT-5 (mean rank difference: Neutral +0.283 → Overstated +0.408), with Gemini 3 showing a smaller but consistent inflation pattern, while Sonnet 4.5 remains near zero across styles (Fig. 6). Overall, Test 2 demonstrates that measuring only held-out accuracy can hide serious brittleness: models may appear reasonable on standard evaluation but still be highly sensitive to rhetorical framing, motivating stability-based reporting (flip rates/TCR and directional shift magnitudes) alongside conventional metrics.

C. DEMOGRAPHIC SENSITIVITY UNDER NAME-ONLY VARIANTS (TEST 3)

Test 3 indicates that name-coded demographic cues have a substantially smaller impact than style in this controlled setup. For the fine-tuned transformers, subgroup accuracies are nearly identical across demographic variants, and mean ordinal rank differences are almost unchanged across race-coded and gender-coded names. The dominant effect is instead global miscalibration (systematic overestimation): DistilBERT shows roughly +0.57 mean rank inflation and RoBERTa roughly +0.74, with negligible gaps between groups. The extreme intersectional comparison (Caucasian Male vs. African-American Female) similarly shows near-indistinguishable shifts for both models (Fig. 8), reinforcing that the main behavior here is global upward calibration error rather than demographic-specific directional bias. For zero-shot LLMs, calibration is again provider-dependent and generally stable under name-only swaps. Sonnet 4.5 remains near-zero and demographically invariant, Gemini 3 shows moderate but consistent upward bias with minimal group differences, and GPT-5 exhibits the largest overall upward shift with a small separation in the extreme comparison (Fig. 9). However, these demographic gaps remain small relative to each model's overall tendency to overestimate. Taken together, Tests 2 and 3 suggest that, under controlled edits, style-driven perturbations dominate the observed instability, while name-only demographic signals produce comparatively minor effects. This highlights the need to evaluate robustness and fairness with carefully controlled counterfactuals: without controlling for writing style, apparent demographic differences may be confounded by presentation effects rather than demographic sensitivity

VI. LIMITATIONS

This study has several limitations. First, the robustness and fairness analyses (Tests 2-3) rely on controlled synthetic perturbations style rewrites and name-coded demographic variants which may not fully reflect real-world resume variation (formatting, omissions, different writing conventions, or richer demographic cues). Second, the controlled evaluations are conducted on a limited sample size (3N inputs from N base resumes, with N = 120), which constrains statistical confidence, especially for subgroup comparisons and extreme-case contrasts. Third, Test 3 uses name-coded proxies for race and gender rather than ground-truth demographic attributes; therefore, the results should be interpreted as sensitivity to surface-level name cues, not definitive real-world fairness outcomes. Fourth, resource constraints limited the experimental scope: we did not fine-tune larger foundation models, run extensive hyperparameter searches, or explore additional prompting regimes (e.g., few-shot prompts) due to computing and API budget limitations. Finally, the TF–IDF + Random Forest baseline appears partly dependent on dataset-specific artifacts, as removing the explicit "total years of experience" feature substantially reduces performance, indicating reduced robustness when duration information is missing or represented differently.

VII. FUTURE WORK

Future work can extend this study in several directions. First, we will broaden controlled evaluations by expanding the style-robustness and name-only demographic sensitivity protocols to larger and more diverse resume sets, enabling stronger statistical confidence and more granular subgroup analyses. Second, we will explore LLM prompting extensions under a strictly standardized protocol (e.g., few-shot exemplars, structured decision rubrics, and constrained output formats) to test whether class balance and calibration improve beyond zero-shot prompting. Third, we will investigate post-hoc calibration and decision thresholding (e.g., temperature scaling, class-specific thresholds, or ordinal-aware calibration) to reduce systematic seniority inflation and improve separation at the Mid boundary. Fourth, we will perform deeper Mid-class error analysis, combining qualitative inspection with targeted feature attribution to identify which signals (e.g., ownership language, scope indicators, impact metrics, leadership cues) most strongly drive between Mid Senior and between Mid Junior confusions, and use these findings to refine data labeling guidelines and model inputs. Finally, we will evaluate model and representation improvements, including

domain-adaptive pretraining on resume/job corpora, alternative encoder backbones, and hybrid approaches that fuse structured numeric signals with contextual text embeddings while enforcing leakage control and consistent evaluation protocols.

VIII. CONCLUSION

This work studied three modeling paradigms for three-class resume seniority prediction (Junior/Mid/Senior) and evaluated them under a unified protocol that separates standard generalization (Test 1) from controlled robustness to rewriting style (Test 2) and controlled sensitivity to name-only demographic cues (Test 3). Under the held-out test set (Test 1), supervised approaches achieved substantially stronger and more balanced performance than prompt-only foundation models. The TF–IDF + Random Forest baseline and fine-tuned Transformer encoders reached the best overall results (≈0.71–0.73 accuracy; ≈0.70–0.73 Macro-F1), while zero-shot LLMs exhibited weaker calibration and class balance (≈0.38–0.44 Macro-F1) with a consistent tendency to over-predict higher seniority. Across all model families, the Mid class remained the dominant error source, with confusion concentrated around the Mid and Senior boundary. This indicates that Mid-level profiles contain mixed or weakly separable signals (partial ownership/impact language without consistently senior scope), making Mid the limiting factor for overall reliability and suggesting that seniority labels encode more than experience duration alone. Controlled evaluations revealed that writing style is a larger driver of instability than demographic name cues in this setup. In Test 2, fine-tuned encoders were sensitive to stylistic rewriting despite fixed facts: both models degraded under Overstated variants, and triplet-level instability was substantial (DistilBERT inconsistency 40.8%; RoBERTa 93.3%). Zero-shot LLMs differed sharply in robustness: GPT-5 showed the highest instability (32.5%) and the strongest upward inflation under Overstated phrasing, Gemini was moderately sensitive (16.7%), and Sonnet 4.5 was highly stable (4.2%) with near-zero average ordinal shift across styles. In Test 3, name-only demographic swaps produced relatively small group differences compared to each model's overall calibration tendencies. Fine-tuned encoders showed strong global upward rank inflation that was similar across race- and gender-coded variants, while LLMs were largely invariant in mean shift (Sonnet near zero; Gemini modest positive; GPT-5 the largest but still small group gap even in the extreme intersectional comparison). Overall, the results support three takeaways: dataset-aligned supervision remains critical for accurate and class-balanced seniority prediction, robustness cannot be inferred from Test 1 accuracy alone, controlled style tests expose substantial sensitivity to self-presentation, and in name-only demographic variants, observed effects are dominated more by global calibration behavior than by strong demographic-specific directional bias. These findings motivate reporting stability and shift-based metrics alongside standard classification scores when deploying resume screening models in practice.

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APPENDIX A
RESULTS TEST 1

A. Baseline (TF-IDF + Random Forest)

The baseline achieves 72.6% accuracy and 0.73 macro-F1with per class F1: Junior =0.84, Mid=0.59, Senior=0.74. As shown in (Fig. 7), most errors occur between Mid and Senior. While junior is classified more reliably. If the total years of experience column is removed, the accuracy drops to 55%, indicating that explicit duration features are a stronger driver of the baseline performance. Confusion matrix for TF-IDF + Random Forest on the test set.

B. Fine-Tuned Transformers (DistilBERT, RoBERTa)

DistilBERT (fine-tuned). We evaluated a fine-tuned DistilBERT classifier for three-class seniority prediction. When training on each table column as a separate input, performance was close to chance (≈50% accuracy). Concatenating all fields into a single unified text input substantially improved performance to 0.726 accuracy and 0.72 Macro-F1. Per-class F1 scores were Junior= 0.90, Mid=0.55, Senior= 0.72, indicating strong performance on Junior and Senior but lower separability for Mid. The dominant error pattern is Mid being confused with Senior (66 Mid to Senior), while Junior is more separable (120 Junior to Junior). RoBERTa (fine-tuned). We evaluated a fine-tuned RoBERTa classifier for three-class seniority prediction. Like DistilBERT, training on individual columns as separate inputs produced near-chance performance (≈50% accuracy), while concatenating all fields into a single unified text substantially improved results to 0.707 accuracy and 0.70 Macro-F1. Per-class F1 scores were Junior=0.87, Mid=0.52, Senior=0.71, showing strong recognition of Junior and Senior but weaker separability for Mid, which is also reflected in the error distribution. Errors are again concentrated in the Mid vs Senior boundary (79 Mid to Senior), while Senior is mostly correct (129 Senior to Senior) and Junior remains relatively stable (109 Junior to Junior).

C. Zero-Shot LLMs (GPT-5, Gemini 3 Pro, Claude Sonnet 4.5)

GPT-5 (zero-shot) is evaluated using a fixed prompt that instructs the model to output exactly one label (Junior/Mid/Senior) given the same resume representation used across models. In this setting, GPT-5 achieves 0.459 accuracy and 0.403 macro-F1, with per-class F1 of Junior=0.296, Mid= 0.241, Senior=0.672. The confusion matrix reveals a strong tendency to overestimate seniority: most Junior resumes are predicted as Mid (91), and most Mid resumes are predicted as Senior (105), while Senior cases are largely predicted correctly (136 correct Senior predictions). This pattern suggests that GPT-5 captures the ordinal progression of seniority (Junior to Mid to Senior) but applies a high threshold for assigning the Junior label, leading to weak Junior and Mid performance. Claude Sonnet 4.5 achieves 0.450 accuracy and 0.385 macro-F1, with per-class F1: Junior=0.418, Mid=0.111, Senior= 0.626. The model exhibits a strong upward bias in its predictions: many Junior samples are pushed to Mid (64 Junior to Mid) and many Mid samples are pushed to Senior (128 Mid to Senior). At the same time, Senior is classified perfectly (140 Senior to Senior), indicating that Claude captures the general seniority direction but struggles to separate early and mid-career profiles. Gemini 3 Pro achieves 0.500 accuracy and 0.435 macro-F1, with per-class F1: Junior=0.130, Mid=0.489, Senior=0.687. The model rarely predicts the Junior class correctly (10 Junior to Junior), instead shifting most Junior resumes to Mid (104 Junior to Mid). Performance is stronger for Mid (88 Mid to Mid) and Senior (112 Senior to Senior), indicating that Gemini captures the overall ordering of seniority but uses a decision threshold that effectively compresses early-career profiles into the Mid-range. Errors are concentrated in Junior to Mid and Mid to Senior confusions.

APPENDIX B
ADDITIONAL SUBGROUP RESULTS FOR TEST 3 (NAME-ONLY DEMOGRAPHIC VARIANTS)

This appendix provides the additional subgroup breakdowns for Test 3 that are omitted from the main Results for brevity. Recall that Test 3 isolates demographic cue effects by swapping name-coded race × gender cues while holding resume content fixed.

A. Fine-tuned Transformers: Race-only and Gender-only Breakdowns

Fig. 10 and Fig. 11 provide the race-only and gender-only subgroup results for the fine-tuned transformer models (DistilBERT and RoBERTa), using the same definition of Δ. These plots support the interpretation in the main text that any observed effect is dominated by global calibration tendencies (overall upward shift) rather than consistent demographic-driven directional bias.

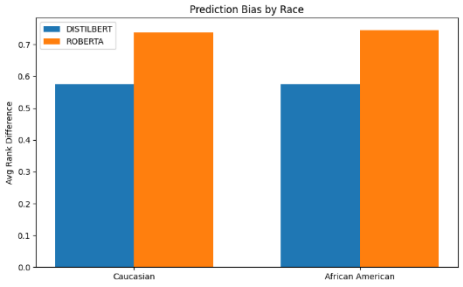

FIGURE 10. Mean ordinal rank shift $\bar{\Delta}$ under race-coded name variants for fine-tuned transformers (DistilBERT, RoBERTa). Positive values indicate upward seniority shifts relative to the neutral prediction.

Both DistilBERT and RoBERTa show nearly identical mean shifts across race-coded groups (Caucasian vs African American), indicating minimal differential sensitivity to race-coded names in this setup. However, the magnitude of the shift is substantial (especially for RoBERTa), suggesting the dominant phenomenon is global miscalibration / overestimation rather than group-specific bias.

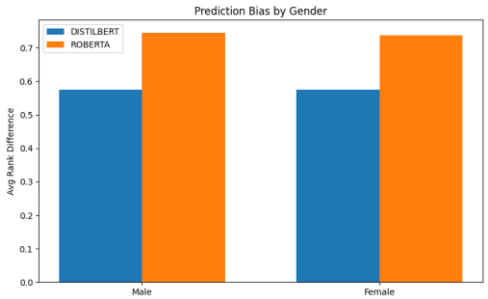

FIGURE 11. Mean ordinal rank shift $\bar{\Delta}$ under gender-coded name variants for fine-tuned transformers (DistilBERT, RoBERTa). Positive values indicate upward seniority shifts relative to the neutral prediction.

The male vs female bars are almost indistinguishable for each transformer model, again indicating low demographic separability under name-only gender cues. Similar to the race-coded plot, the main effect is consistent upward inflation (RoBERTa > DistilBERT), implying that model calibration dominates the behavior more than the demographic cue itself.

B. Zero-shot LLMs: Gender-only Breakdown

While the main Results present the key race-coded and extreme intersectional comparisons for LLMs (Figs. 7 and 9), Fig. 12 provides the remaining gender-only subgroup plot for the zero-shot LLMs (GPT-5, Gemini 3, Sonnet 4.5). This figure complements the main Results by showing whether gender-coded name swaps introduce consistent directional shifts beyond each model's overall calibration.

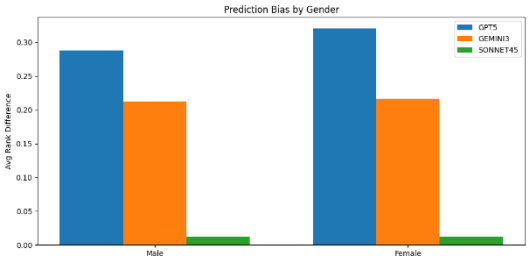

FIGURE 12. Mean ordinal rank shift $\bar{\Delta}$ under gender-coded name variants for zero-shot LLMs (GPT-5, Gemini 3, Sonnet 4.5). Positive values indicate upward seniority shifts relative to the neutral prediction.

Provider differences dominate: GPT-5 shows the largest upward shift, Gemini 3 shows a moderate upward shift, and Sonnet 4.5 remains near zero (high calibration/invariance). The male vs female differences are small relative to each model's overall shift, suggesting limited demographic sensitivity under name-only gender swaps, compared to the much stronger effects observed under controlled style changes in Test 2.

APPENDIX C
RESULTS (TEST 3: DEMOGRAPHIC NAME-ONLY VARIANTS)

This appendix reports subgroup accuracy for Test 3 under name-only demographic variants, complementing the main-text shift analyses (Figs. 7-9) and the additional mean-shift subgroup plots (Appendix B). While the main Results emphasize ordinal rank shift (Δ) to detect demographic sensitivity, accuracy-by-group is included here to verify whether name-only swaps materially change correctness across the four intersectional variants (Caucasian Male, Caucasian Female, African American Male, African American Female). Fine-tuned transformers - Fig. 13 shows that subgroup accuracies for DistilBERT and RoBERTa are tightly clustered across all four intersectional groups, supporting the interpretation in the main text that name-only demographic cues do not meaningfully change correctness. Any remaining differences are small relative to each model's overall error level, indicating that the dominant effect is calibration/overall misclassification rather than group-dependent accuracy shifts.

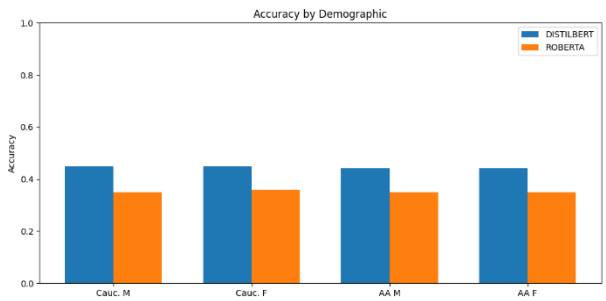

FIGURE 13. Accuracy by intersectional name-only demographic subgroup (Test 3) for fine-tuned transformers (DistilBERT, RoBERTa): Caucasian Male, Caucasian Female, African American Male, African American Female. Subgroup accuracies are closely clustered, indicating minimal demographic-driven accuracy differences under name-only swaps.

Zero-shot LLMs. Fig. 14 reports the same breakdown for GPT-5, Gemini 3, and Sonnet 4.5. As in the main Results, provider-level differences dominate (Sonnet 4.5 highest and most invariant; Gemini moderate; GPT-5 lower and more variable), while subgroup gaps remain small relative to each model's overall accuracy. Together, these plots reinforce that, in this controlled setup, demographic name cues contribute far less to performance variation than model-family calibration tendencies and style effects (Test 2).

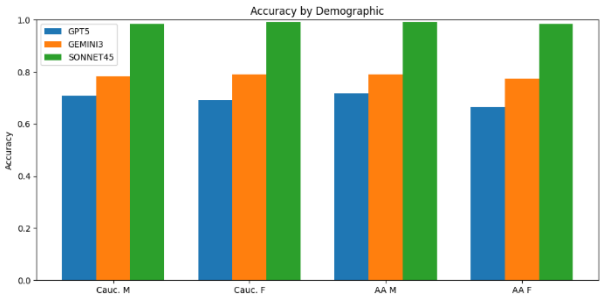

FIGURE 14. Accuracy by name-coded demographic group for the tested LLMs. Sonnet 4.5 remains near-perfect across all groups; GPT-5 shows the largest (though still modest) demographic variation.

E. Fabrikant is a final-year B.Sc. candidate in Data Science at Afeka College of Engineering, Tel Aviv, Israel (expected 2026). He applies machine learning and business intelligence to real-world problems, building data pipelines and analytics workflows using Python, SQL, and Tableau. He is strongly focused on building practical, end-to-end systems, turning ideas into working pipelines from data preparation and modeling to evaluation and deployment. He enjoys hands-on implementation and engineering solutions that can be measured and used in real settings. He previously served as a Lecturer and Mentor at ORT Singalovski and IDF Basmach (2022-2023), teaching 100+ students in programming and data/ML fundamentals. In 2025, he mentored a team in the DigiTech Creathon, supporting an AI-driven MVP and pitch development; the team advanced to the national finals (Top 7 nationwide). His project work includes an end-to-end BI solution for hotel booking cancellation analysis (ETL, predictive modeling, Tableau) and applied machine learning modeling projects (e.g., Titanic and Ames datasets) emphasizing feature engineering and validation. Mr. Fabrikant's current interests include applied machine learning, NLP and evaluation of model reliability in high-stakes settings. He has been recognized for academic excellence, including ranking 1st/59 in Business Intelligence and receiving top scores in applied ML projects.

R. Deeb is a third-year college student, currently in the later stage of my undergraduate studies. My academic experience has given me a strong foundation in structured problem-solving, research-style thinking, and working through complex material with discipline and consistency.